1 A student is creating a presentation containing images and sound.

The student creates 10 bitmap images using a computer.

Each image has a colour depth of 8-bits and a resolution of 800 × 500 pixels.

i. Define the term pixel.

--- **[1]**

ii. Calculate the total file size of the 10 images in kilobytes.

Show your working out.

File size: ……………………………… kilobytes

[2]

iii. State the minimum number of bits that will be needed to represent 240 different colours.

--- **[1]**

2 Binary numbers can also represent images.

The table shows the colours that are used in an image and the binary value for each colour.

Colour	Binary value
Red	0000
Green	0010
Blue	1000
Purple	0110

The metadata states that the image is 3 pixels wide by 4 pixels high.

The data in the file starts in the top left of the image and goes from left-to-right, top-tobottom.

i. State what is meant by **metadata** in an image file.

[1]

ii. The binary data stored for the image is given:

000000000110100000101000011001100110000000101000

A grid is given for the image. Each square is one pixel.

Write the name of the colour in each square that the pixel will show for this image.

[2]

iii. A colour depth of 4 is used. This means 4 bits are used to store the colour for each pixel.

State the maximum number of different colours that can be represented in 4-bits.

[1]

iv. The colour depth is increased to 2 bytes.

State **two** effects that this change can have on the image.

1

2

[2]

3 A student takes a photograph of their science experiment. The image file includes metadata.

Identify **three** pieces of metadata that is often stored with an image.

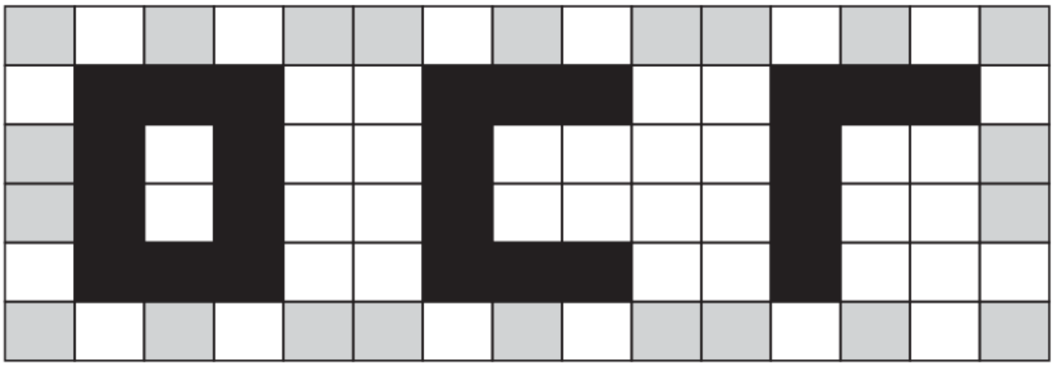
1

2

3

[3]

4(a) The following logo is stored as a bitmap image. Each box represents one pixel, with three different colours being used in the image.



State what is meant by the term image resolution.

[1]

(b) Calculate the fewest number of bits that could be used to store the logo as a bitmap image. You must show your working.

--- **[4]**

(c) Give **two** ways that the file size of the image could be reduced.

1 ---

---

2 ---

---

[2]

(d) Metadata is sometimes stored alongside images.

i. State what is meant by the term metadata.

--- **[1]**

ii. Give **one** example of metadata that could be stored alongside the logo.

--- **[1]**

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1		i	1 mark from • The **smallest unit/part** of an image • A (single) square/dot/diode which has **one/single colour**	1	Accept screen in place of image. BOD block/particle etc. in place of square. <u>Examiner's Comments</u> Some candidates found it challenging to define the term pixel. The most common accurate responses identified it as the smallest part of an image, or a single square of one colour. Some responses described it as a square that contains many colours; while one pixel can have a different colour stored in it at different times, it can only ever have one colour at a time. Other inaccurate responses included describing it as a bit of data or just a square on a screen; when a square on a screen can be made up of many pixels.
		ii	1 mark for one stage of working: • 800 * 500 (= 400 000) • 400 000 * 10 = 4 000 000 • 4 000 000 * 8 (= 32 000 000 bits) • 32 000 000 / 8 = 4 000 000 • 4 000 000 / 1000 = 4 000 1 mark for answer 4000 kilobytes	2	For MP1 accept 8 * 5 BOD. Accept any method of doing calculations e.g. statements, grids, calculations. Accept division by 1024 instead of 1000. Final answer will be 3906.25 kilobytes. If no answer in the final answer space, look for answer clearly identified in working. <u>Examiner's Comments</u> Candidates were often able to accurately show part of the working, most commonly for multiplying 800 and 500. Fewer candidates were able to complete the calculation to gain the correct response. A common missed element was giving the file size of one image instead of multiplying it by 10 to give the file size of the ten images.
		iii	8	1	Allow calculation that equates to 8 <u>Examiner's Comments</u> There were a mix of responses to this question, with some candidates giving 2^240, 256, etc. Some candidates identified the need for 8 bits but gave the response to the question as 2^8, meaning that there would be 256 bits.

Question			Answer/Indicative content	Marks	Guidance
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
2		i	1 mark: • Data about the data/image/file	1	Question is for a definition, not an example. If the definition is not clear, for example details about the image, information about the image – this is NE, but read the example to see if it clarifies. For example: 'Information about the image, such as the number of pixels' give a BOD. Data could be properties / characteristics. <u>Examiner's Comments</u> This question required a definition of the term metadata and many responses correctly defined it as the data about the image, or the data about the file. Some candidates used information, for example the information about the file, which was not precise enough - but they often carried on with an example that supported their statement and demonstrated their understanding. **Misconception** A common misconception was that metadata identifies the colour of each pixel in the image.

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td></td><td></td><td>ii</td><td>1 mark each:
First row: red red purple
Remainder correct and in correct order
<table><tr><td>red</td><td>red</td><td>purple</td></tr><tr><td>blue</td><td>green</td><td>blue</td></tr><tr><td>purple</td><td>purple</td><td>purple</td></tr><tr><td>red</td><td>green</td><td>blue</td></tr></table></td><td>2</td><td>Ignore case/spelling as long as legible.

If a candidate has completed the table in the incorrect layout e.g. right to left, or bottom to top, then award MP2 as a FT if they have done it all correctly.

<u>Examiner's Comments</u>

This question required candidates to consider the binary number and the binary value for each colour, divide the binary number into groups of 4 bits and match them to the appropriate colour. This was completed accurately by many candidates who were able to match the codes and colours.

The instructions stated that the image starts in the top left, but some candidates started in the bottom right instead.

Some responses did not use the colours provided in the question and created their own colour scheme for the image, commonly just using black and white.</td></tr>
<tr><td></td><td></td><td>iii</td><td>16</td><td>1</td><td>Accept any calculation that equates to 16 i.e. 2^4

<u>Examiner's Comments</u>

Candidates were required to calculate the number of different colours that can be represented in 4-bits. This was done by working out how many binary numbers can be created using 4-bits. A common error was giving four colours, or in some cases only one or two colours.</td></tr>
</table>

Question			Answer/Indicative content	Marks	Guidance
		iv	1 mark each to max 2: • The **quality** of the image can be improved • The **file** size will increase // takes up more **storage** space // image has/requires/takes up more data • The number of colours that can be represented/used will increase // BOD more colourful	2	Do not award higher resolution, image size increases, clearer image (NBOD) more detailed image (NBOD). Closer to original is NE on its own because there is not an original image in this context. Mark first answer in each answer space. <u>Examiner's Comments</u> This question was often answered well. Candidates commonly identified that the file size will increase with an increase in colour depth. Some responses also identified that this increase would allow the image to use more colours. ? Misconception A common misconception is that colour depth increases the resolution of the image. This would need to be an increase in the resolution (the number of pixels) as opposed to the number of bits per pixel.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
3			1 mark each to max3 e.g. • Height • Width • Colour/bit depth • Date • Geolocation • File size • File type • Compression type • Author	3	Accept anything reasonable but not features of image e.g. names of people Award resolution for height or width, but max 2 for resolution/dimensions/image size, height, width. 'Colour' on its own is NE. 'Size' on its own is NE. Needs to be what is stored, e.g. date is stored, age of image is not stored. <u>Examiner's Comments</u> This question was answered well with many candidates identifying examples of metadata for an image. There were a wide range of correct responses. The most common responses were the bit or colour depth, the resolution and the file size. Some candidates focused on the device that took the image originally, for example the type of camera, the GPS location or the time and date it was taken.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
4	a		• Number of pixels (in an image) • Height <u>and</u> width (of an image)	1 AO2 1b(1)	Accept pixels per inch / mm / unit area (density)
	b		• 90 (pixels in an image) // 15 x 6 (pixels in image) • Multiply pixels x bits per pixel • …2 bits required per pixel (because 3 colours) • 180 bits overall answer	4 AO1 1b(2) AO1 1b(2)	Must clearly show multiplication for 3rd BP
	c		• Reduce number of pixels / resolution • Reduce number of colours • Use lossy compression • Use lossless compression	2 AO2 1a(2)	Accept descriptive answers linked to given logo (e.g "change to black and white only") "Make image smaller" is NE Allow compression by itself for one answer.
	d	i	• Data <u>about</u> data / the image/file // properties of the file	1 AO1 1b(2)	Do not accept examples without a definition.
		ii	e.g. • height • width • colour depth • resolution • geolocation • date/time created/last edited // timestamp • file type • author details	1 AO1 1a(2)	Accept any sensible data that could be stored alongside an image. Do not accept filename
			Total	**9**	